Similarly we find that $(g \circ f)(2) = 1$ and $(g \circ f)(3) = 2$. So $g \circ f = \{(1,1),(2,1),(3,2)\}$. Similarly $f \circ g = \{(2,2),(3,2),(4,2)\}$.

Example 1.26 Let $f = \{(1,4),(2,5),(3,5)\}$ and $g = \{(4,1),(5,2),(6,4)\}$. Find $g \circ f$. Can you find $f \circ g$?

Solution:
Clearly, $g \circ f = \{(1,1),(2,2),(3,2)\}$. But $f \circ g$ is not defined because the range of $g = \{1,2,4\}$ is not contained in the domain of $f = \{1,2,3\}$.

Example 1.27 Let f and g be the two functions from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = 3x - 4$ and $g(x) = x^2 + 3$. Find $g \circ f$ and $f \circ g$.

Solution:
We have,
$(g \circ f)(x) = g(f(x)) = g(3x-4) = (3x-4)^2 + 3 = 9x^2 - 24x + 19.$
$(f \circ g)(x) = f(g(x)) = f(x^2+3) = 3(x^2+3) - 4 = 3x^2 + 5.$



Here we have $f \circ g \neq g \circ f$. Thus the operation "composition of functions" is in general not commutative.

Theorem 1.2: Let $f : A \to B$ and $g : B \to C$ be two functions. If f and g are one-to-one, then $g \circ f$ is one-to-one.

Proof. Let $x \neq y$ in A. Since f is one-to-one, $f(x) \neq f(y)$. Since g is one-to-one, $g(f(x)) \neq g(f(y))$. That is, $x \neq y \Rightarrow (g \circ f)(x) \neq (f \circ g)(y)$. Hence $g \circ f$ is one-to-one. ■

Example 1.28 Show that the statement,

"**if f and $g \circ f$ are one-to-one, then g is one-to-one**" is not true.

Solution:
To claim a statement is not true we have to prove by giving one counter example. Consider the diagram given in Figure 1.40.

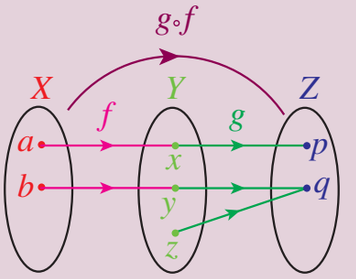


Figure 1.40

Clearly f and $g \circ f$ are one-to-one. But g is not one-to-one. Thus from the above diagram it shows that the statement is not true.